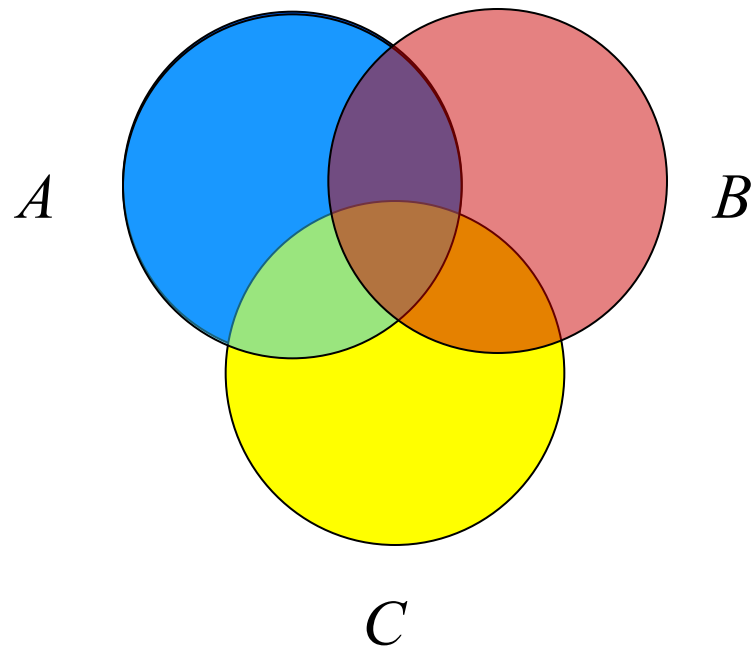


# Set Theory



# This Lecture

We will first introduce set theory before we do counting.

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

# Sets

A **set** is a collection of mathematical objects,  
with the collection treated as a single mathematical object.

- Examples:
- real numbers,  $\mathbb{R}$
  - complex numbers,  $\mathbb{C}$
  - integers,  $\mathbb{Z}$
  - empty set,  $\emptyset$

# Defining Sets

A set is just a collection of objects.

Sets can be defined directly:

e.g.  $\{1, 2, 4, 8, 16, 32, \dots\}$ ,  
 $\{\text{CSC1130}, \text{CSC2110}, \dots\}$

Order, number of occurrence are not important.

e.g.  $\{A, B, C\} = \{C, B, A\} = \{A, A, B, C, B\}$

A set can be an element of another set.

$\{1, \{2\}, \{3, \{4\}\}\}$

# Defining Sets by Predicates

The set of elements,  $x$ , in  $A$  such that  $P(x)$  is true.

$$\{x \in A \mid P(x)\}$$

$$\{-1, 0, 1, 2, 3, 4\} = \{x \in \mathbb{Z} \mid -2 < x < 5\}$$

$$\{x \in \mathbb{R} \mid -2 < x < 5\}$$

The set of prime numbers:

$$\{p \in \mathbb{Z}^+ \mid (p > 1) \wedge (\forall a \in \mathbb{Z} \forall b \in \mathbb{Z} (a > 1 \wedge b > 1) \rightarrow a \cdot b \neq p)\}$$

# Membership

The most important question in set theory is whether an element is in a set.

$$x \in A$$

$x$  is an **element** of  $A$

$x$  is **in**  $A$

Examples:

$$\pi/2 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$\pi/3 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$14/2 \in \{7, \text{"Albert"}, \pi/2, \mathbf{T}\}$$

$$7 \notin$$

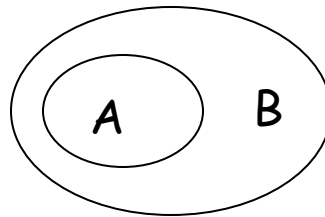
$$2/3 \notin$$

# Containment

$A \subseteq B$        $A$  is a **subset** of  $B$   
 $A$  is **contained in**  $B$

Every element of  $A$  is also an element of  $B$ .

$A \subseteq B \iff \forall x, \text{ if } x \in A \text{ then } x \in B$



$A \not\subseteq B \iff \exists x, x \in A \text{ and } x \notin B$

$A \subset B$        $A$  is a **proper subset** of  $B$

$A = B \iff A \subseteq B \text{ and } B \subseteq A$

# Examples

$\emptyset \in \mathbb{R}$ ,

$$\{3\} \subset \{5, 7, 3\}$$

$\emptyset \in$  every set,

$$A \in A$$

$$\text{prime} \not\subseteq \text{odd}$$

$$\{a, \{b\}, c\} \not\subseteq \{a, b, c\}$$

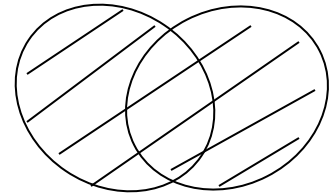
$$\{\{b\}\} \neq \{b\}$$

# This Lecture

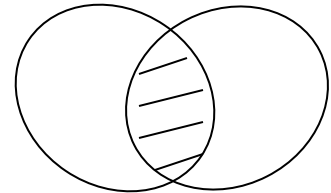
- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

# Basic Operations on Sets

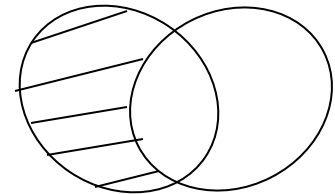
union:  $A \cup B ::= \{x \mid (x \in A) \vee (x \in B)\}$



intersection:  $A \cap B ::= \{x \mid x \in A \wedge x \in B\}$

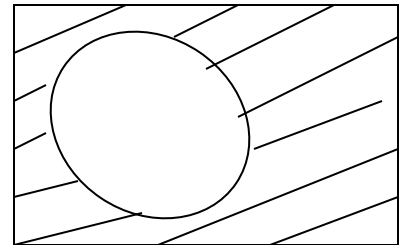


difference:  $A - B ::= \{x \mid (x \in A) \wedge (x \notin B)\}$



complement:  $\overline{A} ::= \{x \in D \mid x \notin A\} = D - A$

Complement can be defined only if the domain is defined



## Examples

$$A = \{1, 3, 6, 8, 10\} \quad B = \{2, 4, 6, 7, 10\}$$

$$A \cap B = \{6, 10\}, \quad A \cup B = \{1, 2, 3, 4, 6, 7, 8, 10\} \quad A - B = \{1, 3, 8\}$$

$$\text{prime} \cap \text{even} = \{2\}, \quad \text{even} \cap \text{odd} = \emptyset$$

$$A = \{x \mid x = 2k \text{ for some integer } k\}, \quad B = \{x \mid x = 3k \text{ for some integer } k\}$$

$$A \cap B = \{x \mid x = 6k \text{ for some integer } k\}$$

$$A \cup B = \{x \mid x \text{ is either a multiple of 2 or a multiple of 3 (or both)}\}$$

$$A - B = \{x \mid x \text{ is even but not a multiple of 3}\}$$

Let  $D$  (i.e. the domain) be the set of integers, then

$\overline{A}$  is the set of odd numbers

# Partitions of Sets

Two sets are **disjoint** if their intersection is empty.

A collection of nonempty sets  $\{A_1, A_2, \dots, A_n\}$  is a **partition** of a set  $A$  if and only if

$$A = A_1 \cup A_2 \cup \dots \cup A_n$$

and  $A_1, A_2, \dots, A_n$  are **mutually disjoint**.

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$$a \equiv b \pmod{n}$$

For a positive integer  $n$ , two integers  $a$  and  $b$  are said to be **congruent modulo  $n$** , if their difference  $a - b$  is an integer multiple of  $n$ . The number  $n$  is called the **modulus** of the congruence.

$38 \equiv 14 \pmod{12}$  congruence of  $a$  and  $b$  can also be thought of as asserting that these two numbers have the same remainder after dividing by the modulus  $n$

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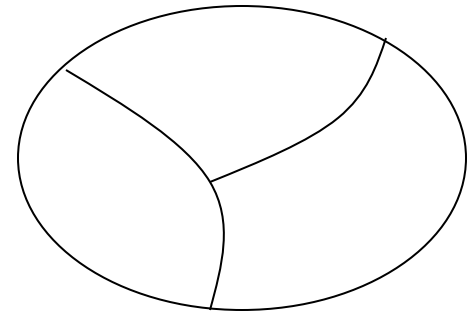
Example. Let  $A$  be the set of integers.

$A_1$  be the set of integers  $\equiv 1 \pmod{3}$

$A_2$  be the set of integers  $\equiv 2 \pmod{3}$

$A_3$  be the set of integers  $\equiv 0 \pmod{3}$

$\{A_1, A_2, A_3\}$  is a partition of  $A$



# Power Sets

power set:  $\text{pow}(A) ::= \{S \mid S \subseteq A\}$

$$\text{pow}(\{a,b\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$$

$$\text{pow}(\{a,b,c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

$$\begin{aligned} \text{pow}(\{a,b,c,d\}) = & \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \\ & \{a,b\}, \{a,c\}, \{b,c\}, \{a,d\}, \{b,d\}, \{c,d\}, \\ & \{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}, \{a,b,c,d\}\} \end{aligned}$$

If  $A$  has  $n$  elements, then the  $\text{pow}(A)$  has  $2^n$  elements.

$P(n)$  = If  $A$  has  $n$  elements, then the  $\text{pow}(A)$  has  $2^n$  elements.

Basis:  $P(0)$  is true, as the set with 0 elements has 1 subset (itself) and  $2^0 = 1$ .

Inductive step: Prove  $P(k) \rightarrow P(k+1)$

Let  $T$  be a set such that  $|T| = k+1$  and  $T = \{e_1, e_2, \dots, e_k, e_{k+1}\}$ .

Let  $S = \{e_1, e_2, \dots, e_k\}$ . Then  $|S| = k$ , so,  $S$  has  $2^k$  subsets (by induction).

Note that  $T = S \cup \{e_{k+1}\}$ , so every subset of  $S$  is also a subset of  $T$ .

Any subset of  $T$  either contains the element  $e_{k+1}$ , or it doesn't contain  $e_{k+1}$ .  
If a subset of  $T$  doesn't contain  $e_{k+1}$ , then it is also a subset of  $S$ , and there are  $2^k$  of those subsets.

On the other hand, if a subset of  $T$  does contain the element  $e_{k+1}$ , then that subset is formed by including  $e_{k+1}$  in one of the  $2^k$  subsets of  $S$ , so  $T$  has  $2^k$  subsets containing  $e_{k+1}$ .

We have shown that  $T$  has  $2^k$  subsets containing  $e_{k+1}$ , and another  $2^k$  subsets not containing  $e_{k+1}$ , so the total number of subsets of  $T$  is  $2^k + 2^k =$   
(2)  $2^k = 2^{k+1}$

[QED]

# Cartesian Products

Given two sets  $A$  and  $B$ , the Cartesian product  $A \times B$  is the set of all **ordered** pairs  $(a,b)$ , where  $a$  is in  $A$  and  $b$  is in  $B$ .

Ordered pairs means the ordering is important, e.g.  $(1,2) \neq (2,1)$

e.g.  $A = \{x,y\}$   $B = \{1,2,3\}$   
 $A \times B = \{ (x,1), (x,2), (x,3), (y,1), (y,2), (y,3) \}$

If  $A$  has  $n$  elements and  $B$  has  $m$  elements, then  $A \times B$  has  $mn$  elements.

The definition can be generalized to any number of sets.

e.g.  $A = \{1,2\}$ ,  $B = \{x,y\}$ ,  $C = \{T,F\}$   
 $A \times B \times C = \{ (1,x,T), (1,x,F), (1,y,T), (1,y,F),$   
 $(2,x,T), (2,x,F), (2,y,T), (2,y,F) \}$

# This Lecture

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

# Set Identities

Some basic properties of sets

$$A \cap B \subseteq A$$

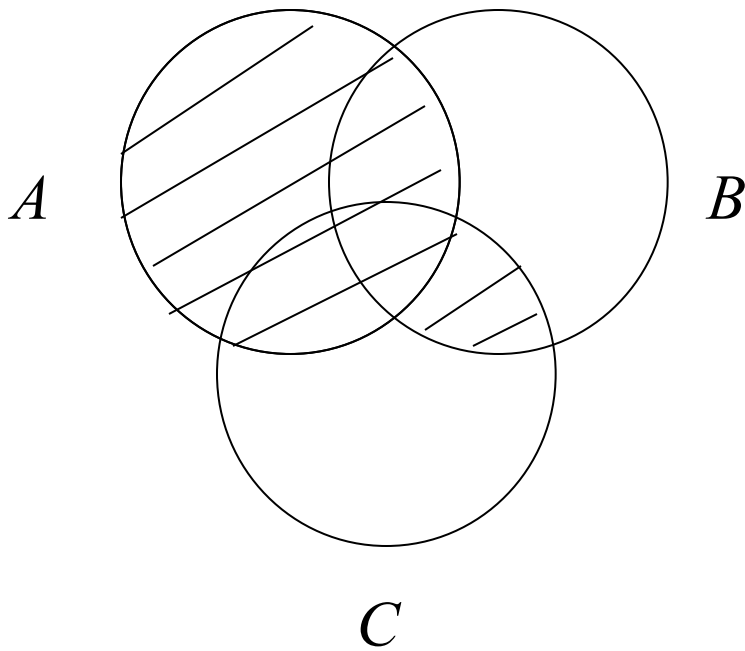
$$A \subseteq A \cup B$$

$$A \subseteq B \text{ and } B \subseteq C \rightarrow A \subseteq C$$

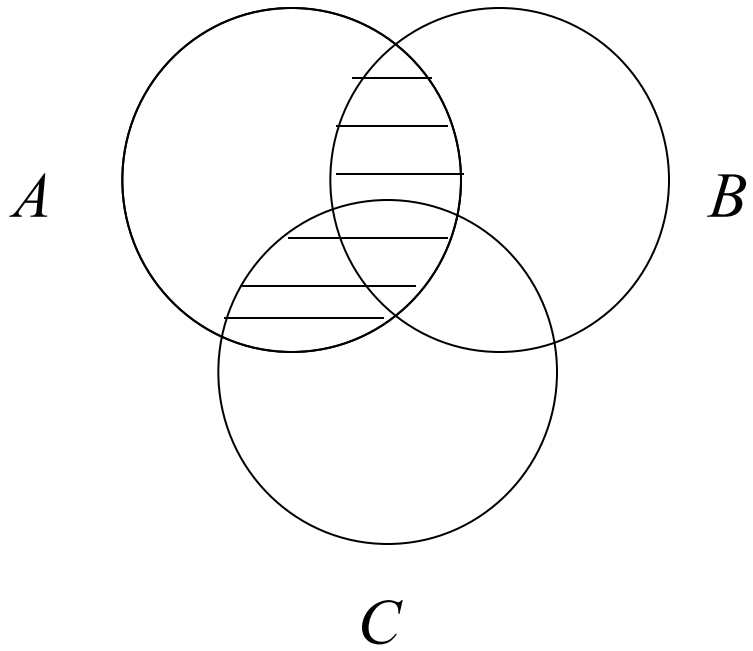
## Set Identities

Distributive Law:  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$  (1)

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$  (2)



(1)

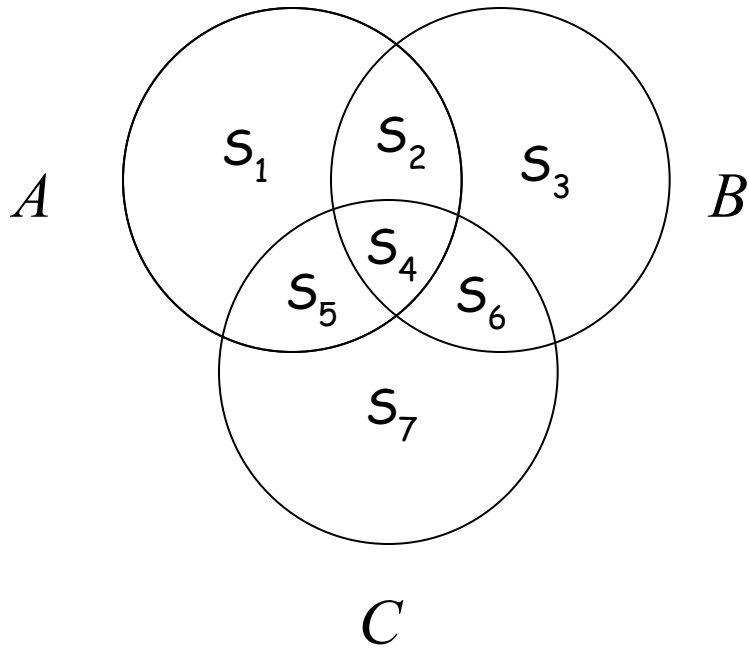


(2)

# Set Identities

Distributive Law:  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

If you insist on proving this law more formally...



$$A = S_1 \cup S_2 \cup S_4 \cup S_5$$

$$B \cap C = S_4 \cup S_6$$

$$A \cup (B \cap C) = S_1 \cup S_2 \cup S_4 \cup S_5 \cup S_6$$

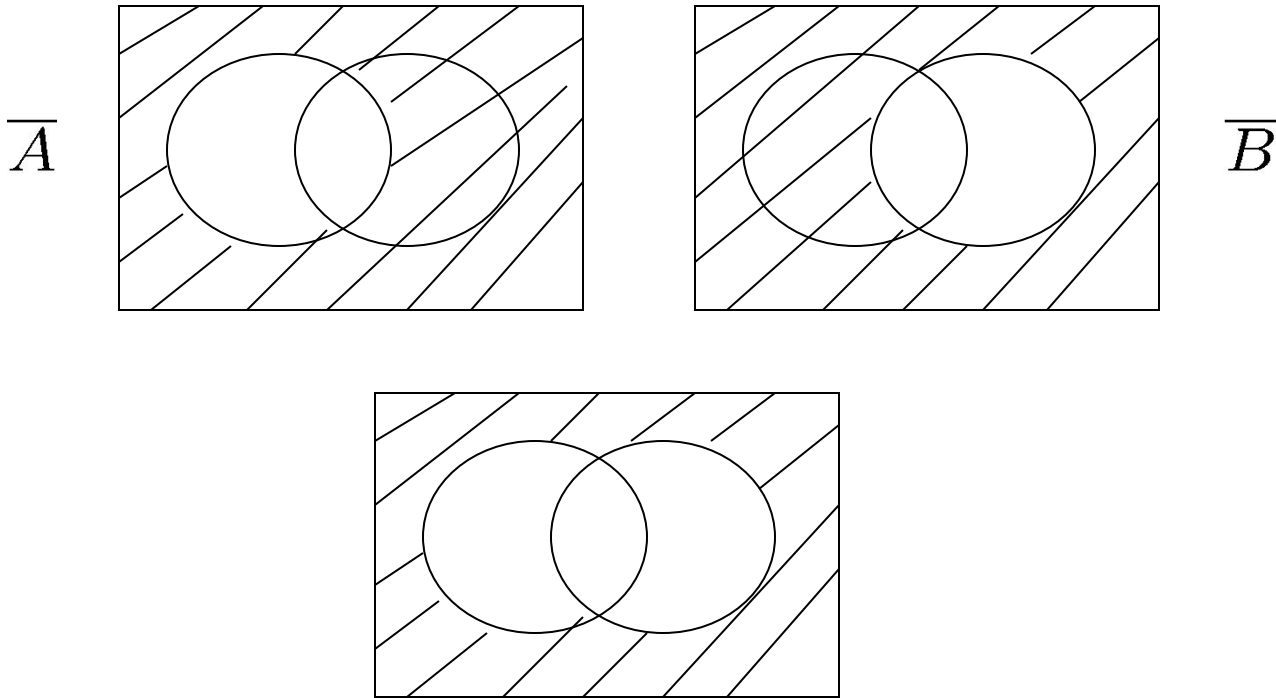
$$(A \cup B) = S_1 \cup S_2 \cup S_3 \cup S_4 \cup S_5 \cup S_6$$

$$(A \cup C) = S_1 \cup S_2 \cup S_4 \cup S_5 \cup S_6 \cup S_7$$

$$(A \cup B) \cap (A \cup C) = S_1 \cup S_2 \cup S_4 \cup S_5 \cup S_6$$

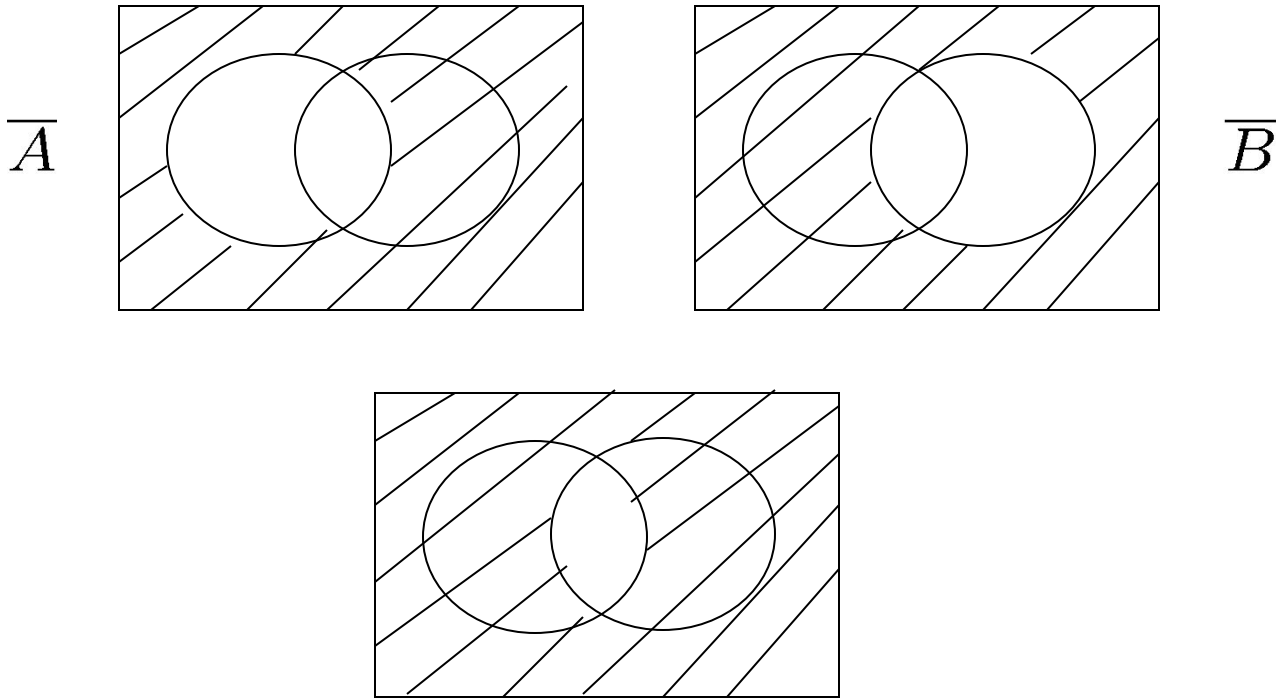
# Set Identities

De Morgan's Law:  $\overline{A \cup B} = \overline{A} \cap \overline{B}$



# Set Identities

De Morgan's Law:  $\overline{A \cap B} = \overline{A} \cup \overline{B}$



## Exercises

$$A - (A \cap B) = A - B?$$

$$(A - B) \cup (B - C) = A - C?$$

$$(A \cup B) - C = (A - C) \cup (B - C)?$$

# This Lecture

- Basic Definitions
- Operations on Sets
- Set Identities
- Russell's Paradox

## Russell's Paradox

Let  $W ::= \{S \in \text{Sets} \mid S \notin S\}$

so  $S \in W \leftrightarrow S \notin S$

|                 |
|-----------------|
| Is $W$ in $W$ ? |
|-----------------|

# Russell's Paradox

- "abnormal" set: if it is a member of itself
- "normal" set: otherwise.
- Example:
  - take the set of all squares  $S$ .
  - $S$  is not itself a square
  - $S$  is "normal".
  - the complementary set that contains all non-squares,  $S'$
  - that set is itself not a square
  - so  $S'$  is "abnormal".

# Russell's Paradox

- Now we consider the set of all normal sets,  $R$ .
- Now, try to determine whether  $R$  is normal or abnormal
- *Case 1:  $R$  is a normal set:*
  - it would be contained in the set of normal sets (itself),
  - and therefore be abnormal;
- *Case 2:  $R$  is abnormal:*
  - it would not be contained in the set of normal sets
  - So, it would not be contained in itself
  - and therefore be normal.
- This leads to the conclusion that  $R$  is neither normal nor abnormal: Russell's paradox.

# Barber's Paradox

There is a male barber who shaves all those men,  
and only those men,  
who do not shave themselves.

Does the barber shave himself?

# This Lecture

Just make sure you understand the basics of set theory.

- Basic Definitions
- Operations on Sets
- Set Identities